



Garry Kasparov playing against Deep Blue, the chess-playing computer built by IBM

the best chess computer at the time – Chess 4.7 – but the challenge of the game made Levy admit that computers would beat humans in chess soon. Sure enough, a few years later, IBM’s Deep Thought managed to beat Levy.

Computer chess programs work with logic that’s essentially based on having a large database. As the available memory or storage space started increasing, it became that much easier for computer scientists to start programming optimal chess play algorithms. Soon, the race turned into beating the clock, since official chess matches are played under a time limit. Scientists started using different types of optimisation techniques, such as board representation (pre-fed chess positions on the board), search techniques (feeding every possible move and programming the computer to search it efficiently) and others. Search offered the most amount of upscalability in this and eventually became the technique used by all the leading chess computers.

But the biggest draw in computer-human chess has been the epic duel between IBM’s Deep Blue and a man many consider to be the greatest chess player of all time, Russia’s Garry Kasparov. In their first match-up in 1996, Kasparov managed to beat Deep Blue convincingly. But the next year, IBM updated Deep Blue – some say, to specifically beat Kasparov by programming it with every game Kasparov has ever